

LAB # 7**IMPLEMENTATION OF WHILE AND DO-WHILE LOOP
STRUCTURE****Lab Tasks**

- 1- Write a program that prints the first 50 even numbers by using while loop.
- 2- Write a program that checks whether a number is prime or not by using if else structure and while loop.
- 3- Write a program to calculate the factorial of any number using while loop.
- 4- Develop a program to perform two simple transactions in a bank as long as user enters y/Y. to continue.

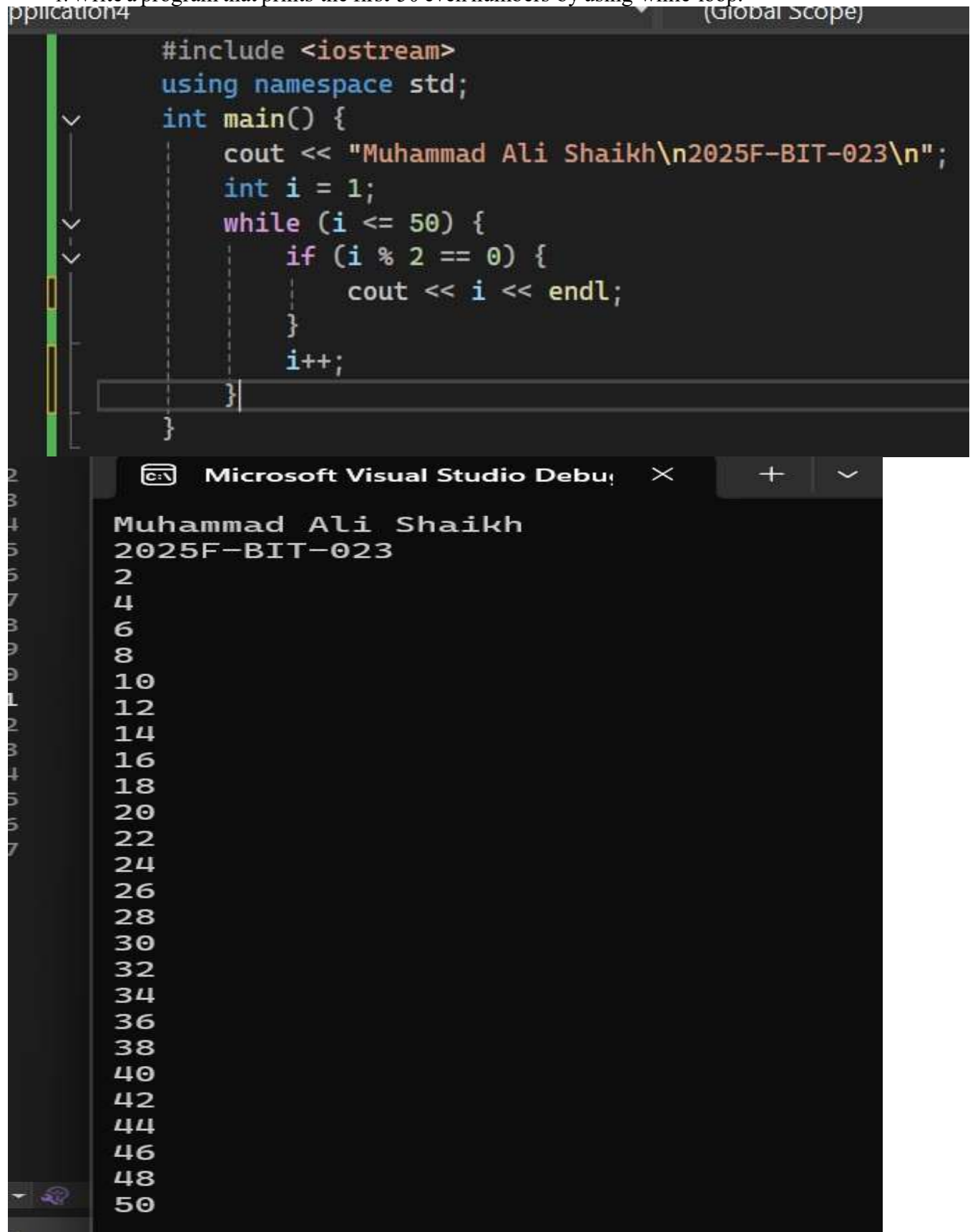
Enter your ID: ****	Main Menu ***** 1. Deposit Money 2. Withdraw Amount 3. Login as Different User Select your choice	<i>(after completing the selected transaction)</i> Do you want to continue? [y/Y] _ <i>(goes to Main Menu, if y/Y is pressed)</i>
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- 5- Write only the necessary statements to print all the numbers in the range 100 - 150 which are divisible by 4 (tab separated), using while loop.
- 6- Write a program that takes a positive integer (<9999) as input and expresses the number in words. Use do-while loop

Input: Enter a positive integer (less than 9999) : 6342

Output: 6 thousand 3 hundred 4 tens and 2 ones

1. Write a program that prints the first 50 even numbers by using while loop.



The image shows a screenshot of a C++ program in Visual Studio. The code is as follows:

```
#include <iostream>
using namespace std;
int main() {
    cout << "Muhammad Ali Shaikh\n2025F-BIT-023\n";
    int i = 1;
    while (i <= 50) {
        if (i % 2 == 0) {
            cout << i << endl;
        }
        i++;
    }
}
```

The output of the program is displayed in the console window, showing the name and ID followed by the first 50 even numbers:

```
Muhammad Ali Shaikh
2025F-BIT-023
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
```

2. Write a program that checks whether a number is prime or not by using if else structure and while loop.

```
onsoleApplication4 (Global Scope)
1  #include <iostream>
2  using namespace std;
3  int main() {
4      cout << "Muhammad Ali Shaikh\n2025F-BIT-023";
5      int num, i = 2;
6      cout << "\nEnter number: ";
7      cin >> num;
8      if (num < 2) {
9          cout << "not a prime number";
10         return 0;
11     }
12     while (i * i <= num) {
13         if (num % i == 0) {
14             cout << "not a prime number";
15             return 0;
16         }
17         i++;
18     }
19     cout << "prime number";
20     return 0;
21 }
```

Microsoft Visual Studio Debug Console

```
Muhammad Ali Shaikh
2025F-BIT-023
Enter number: 11
prime number
```

3. Write a program to calculate the factorial of any number using while loop.

```

Application4 (Global Scope)
#include <iostream>
using namespace std;
int main() {
    cout << "Muhammad Ali Shaikh\n2025F-BIT-023";
    int i=1,num,fac=1;
    cout << "\nEnter number: ";
    cin >> num;
    while (i<= num) {
        fac *= i;
        i++;
    }
    cout <<"factorial of "<<num<<" is " << fac;
}

```

Microsoft Visual Studio Debug Console

```

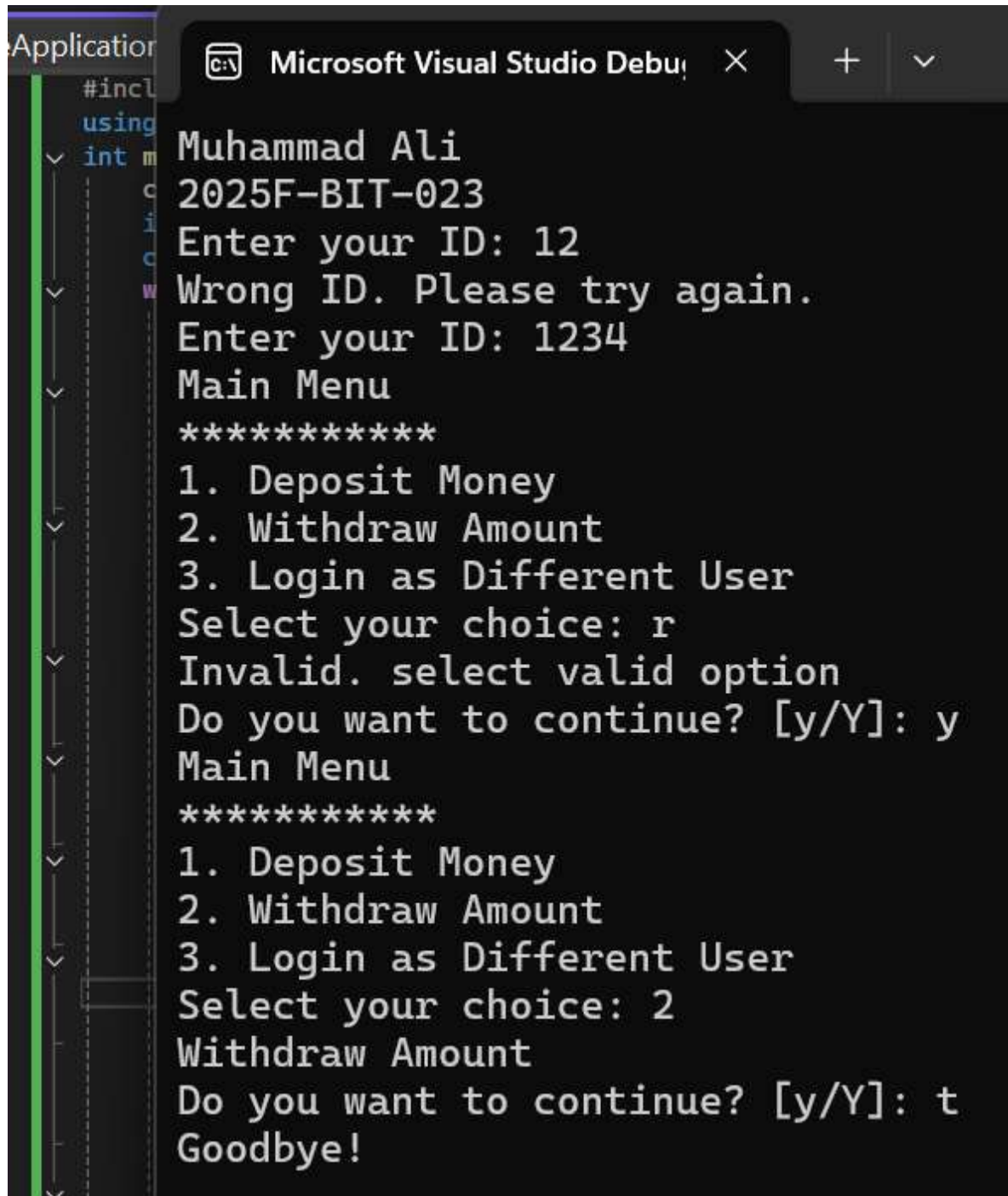
Muhammad Ali Shaikh
2025F-BIT-023
Enter number: 5
factorial of 5 is 120
C:\Users\DELL\source\repos\ConsoleApplication4\Debug\ConsoleApplication4.exe

```

4. Develop a program to perform two simple transactions in a bank as long as user enters y/Y. to continue.

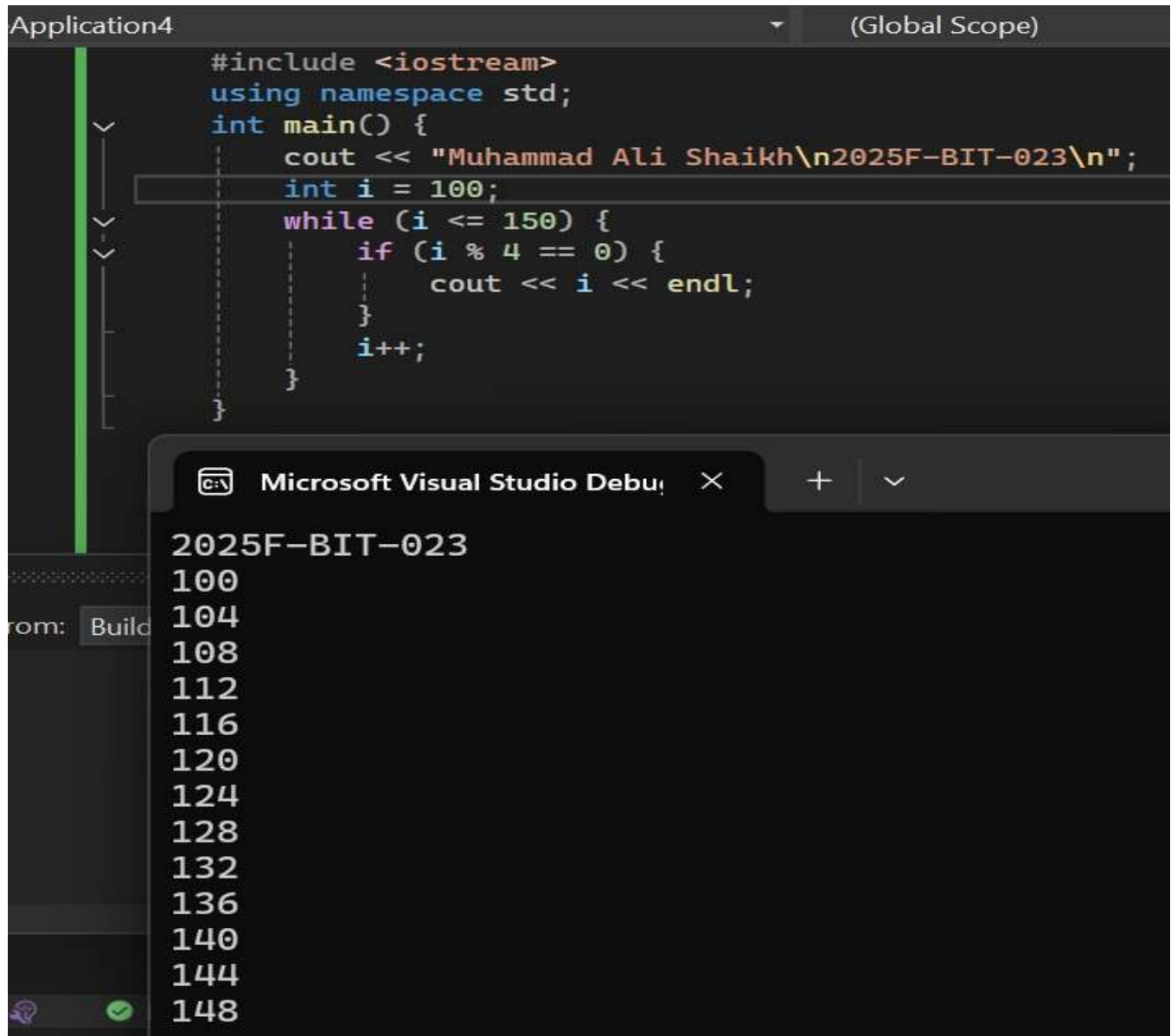
Enter your ID: ****	Main Menu ***** 1. Deposit Money 2. Withdraw Amount 3. Login as Different User Select your choice	<i>(after completing the selected transaction)</i> Do you want to continue? [y/Y] _ <i>(goes to Main Menu, if y/Y is pressed)</i>
---------------------	---	---

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      cout << "Muhammad Ali\n2025F-BIT-023\n";
5      int ID = 1234, id;
6      char choice = 'a';
7      while (choice != 'y' && choice != 'Y') {
8          cout << "Enter your ID: ";
9          cin >> id;
10         if (id != ID) {
11             cout << "Wrong ID. Please try again.\n";
12             continue;
13         }
14         do {
15             cout << "Main Menu\n*****\n1. Deposit Money\n2. Withdraw Amount\n3. Login as Different User\nSelect your choice: ";
16             cin >> choice;
17
18             if (choice == '1') {
19                 cout << "Deposit Money\n";
20             }
21             else if (choice == '2') {
22                 cout << "Withdraw Amount\n";
23             }
24             else if (choice == '3') {
25                 cout << "Login as Different User\n";
26             }
27             else {
28                 cout << "Invalid choice. Please select a valid option.\n";
29             }
30             cout << "Do you want to continue? [y/Y]: ";
31             cin >> choice;
32         } while (choice == 'y' || choice == 'Y');
33
34         if (choice != 'y' && choice != 'Y') {
35             cout << "Goodbye!\n";
36             break;
37         }
38     }
39     return 0;
40 }
```



```
Application  Microsoft Visual Studio Debug Console  X  +  v
#include
using
int m
c
i
c
w
Muhammad Ali
2025F-BIT-023
Enter your ID: 12
Wrong ID. Please try again.
Enter your ID: 1234
Main Menu
*****
1. Deposit Money
2. Withdraw Amount
3. Login as Different User
Select your choice: r
Invalid. select valid option
Do you want to continue? [y/Y]: y
Main Menu
*****
1. Deposit Money
2. Withdraw Amount
3. Login as Different User
Select your choice: 2
Withdraw Amount
Do you want to continue? [y/Y]: t
Goodbye!
```

5. Write only the necessary statements to print all the numbers in the range 100 - 150 which are divisible by 4 (tab separated), using while loop.



The screenshot shows a C++ program in Visual Studio. The code is as follows:

```
#include <iostream>
using namespace std;
int main() {
    cout << "Muhammad Ali Shaikh\n2025F-BIT-023\n";
    int i = 100;
    while (i <= 150) {
        if (i % 4 == 0) {
            cout << i << endl;
        }
        i++;
    }
}
```

The output window shows the following results:

```
2025F-BIT-023
100
104
108
112
116
120
124
128
132
136
140
144
148
```

6. Write a program that takes a positive integer (<9999) as input and expresses the number in words. Use do-while loop

Input: Enter a positive integer (less than 9999) : 6342

Output: 6 thousand 3 hundred 4 tens and 2 ones

The screenshot shows a C++ application window titled 'ConsoleApplication7' with '(Global Scope)' selected. The code is as follows:

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      cout << "Muhammad Ali\n2025F-BIT-023\n";
5      int num;
6      do {
7          cout << "Enter a positive integer (less than 9999): ";
8          cin >> num;
9      } while (num <= 0 || num >= 9999);
10
11     int thousand = num / 1000;
12     int hundred = (num / 100) % 10;
13     int ten = (num / 10) % 10;
14     int one = num % 10;
15     if (thousand > 0) {
16         cout << thousand << " thousand" << endl;
17     }
18     if (hundred > 0) {
19         cout << hundred << " hundred" << endl;
20     }
21     if (ten > 0) {
22         cout << ten << " tens" << endl;
23     }
24     if (one > 0) {
25         cout << one << " ones" << endl;
26     }
27     else {
28         cout << "invalid integer";
29     }
30 }
31
32
33
34
35
36
```

The output window, titled 'Microsoft Visual Studio Debug Console', shows the following execution results:

```
Muhammad Ali
2025F-BIT-023
Enter a positive integer (less than 9999): 4528
4 thousand
5 hundred
2 tens
8 ones
```

At the bottom left, the zoom level is set to 76% and the status bar shows 'Ready'.